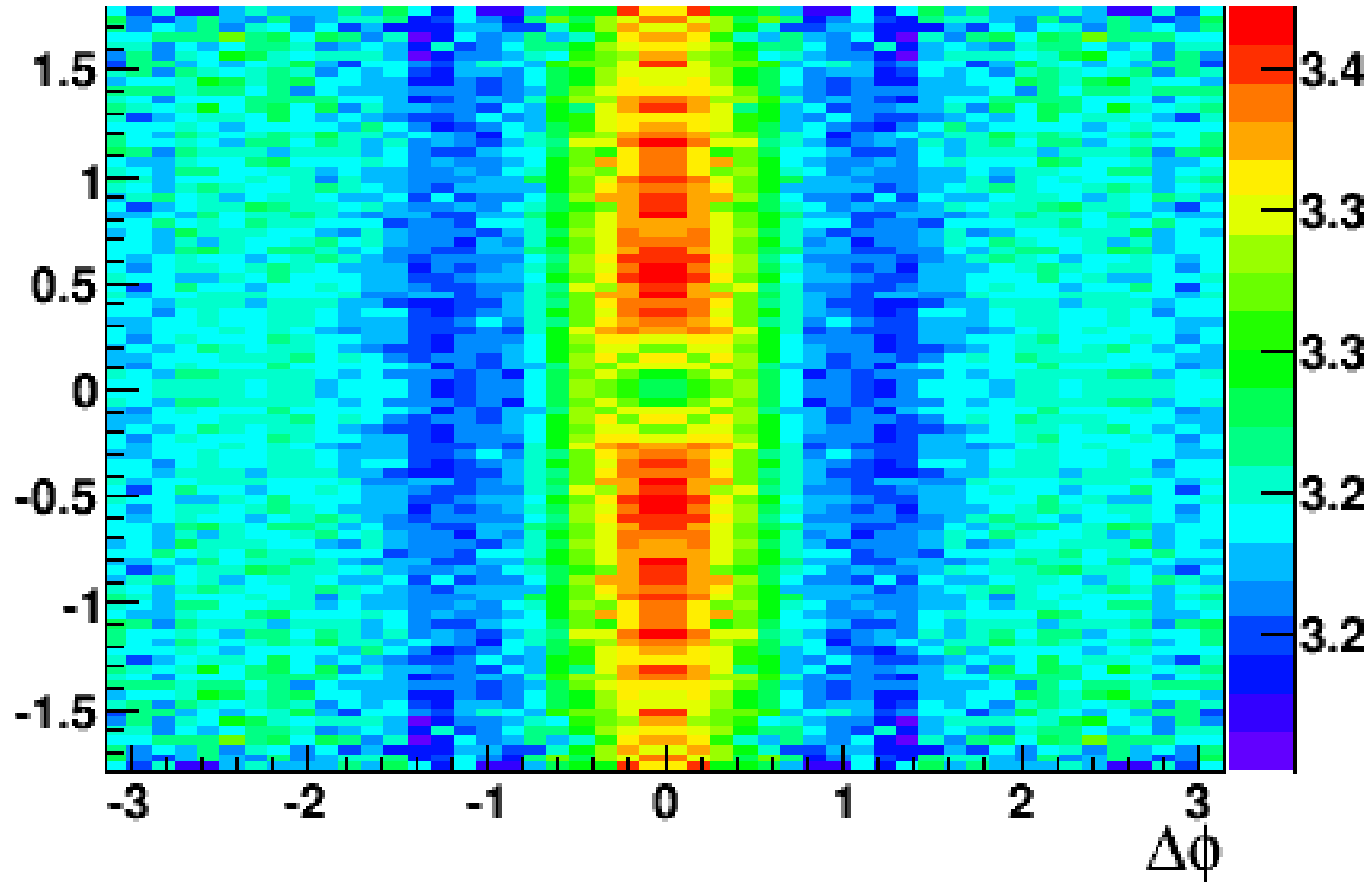


Collision system and energy dependence of di-hadron correlations

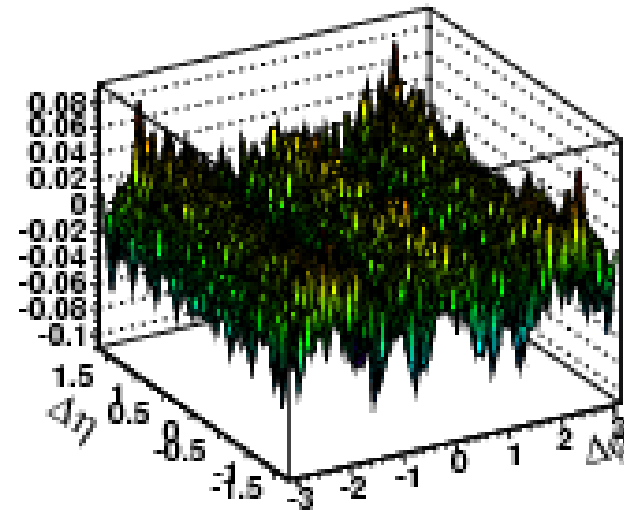
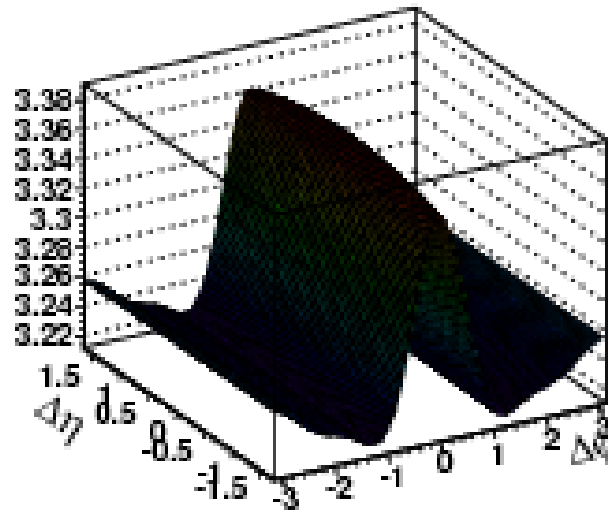
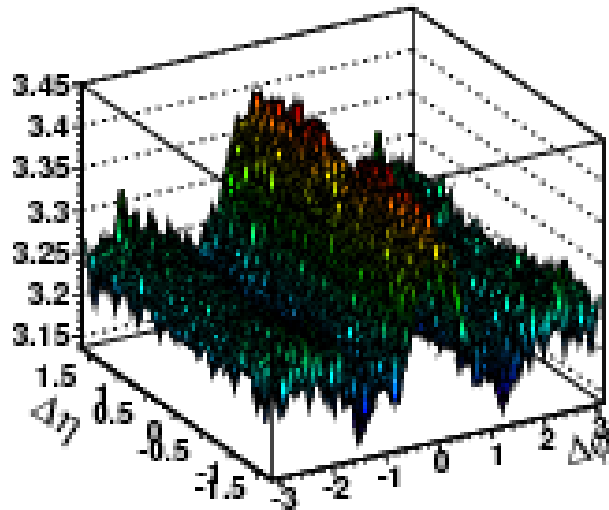
PAs: Christine Nattrass, Jana Bielcikova,
Helen Caines, Jörn Putschke

Trying to subtract d+Au from Au+Au 0-12%



- Subtracting d+Au signal distorts near-side jet-like peak
- This makes sense because the near-side jet-like peak is broader in central Au+Au than d+Au, while the yield is comparable

Trying to subtract d+Au from Au+Au 0-12%



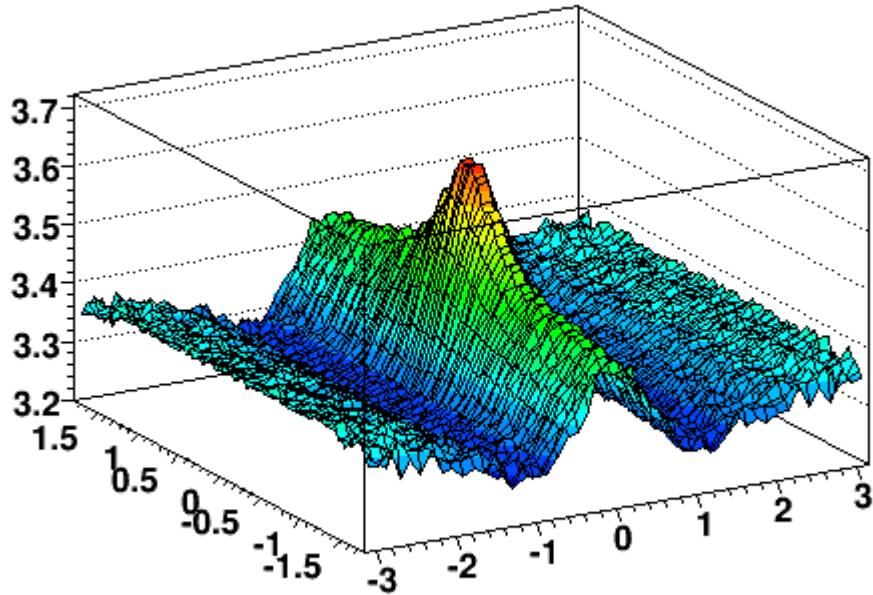
- Residuals look bad

Trying to subtract d+Au from Au+Au 0-12%

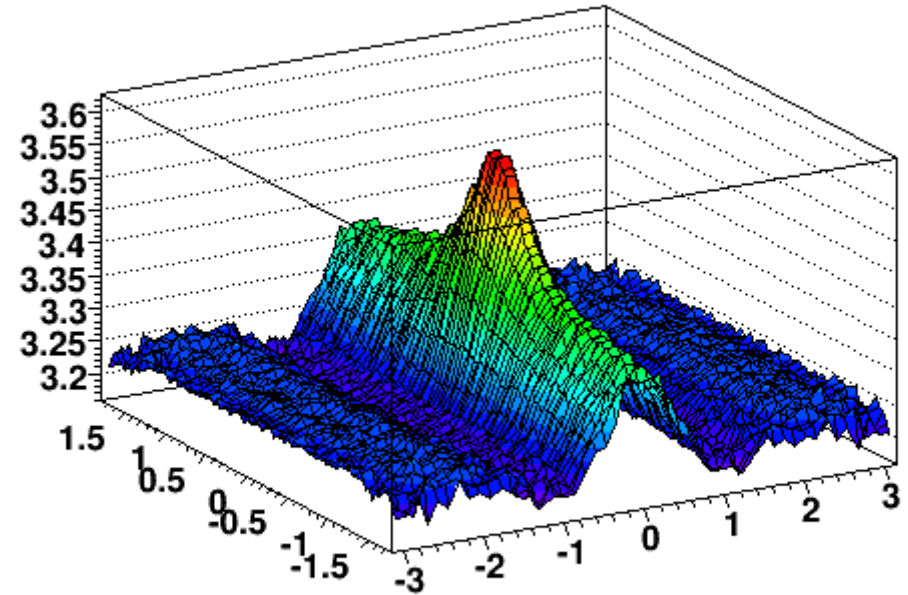
	Standard	After subtracting d+Au
Background	3.34776	3.21886
v1	0.00092	-0.00511
(v2)^2	0.00811	0.00002
(v3)^2	0.00499	0.00285
(v4/v2)^2	0.24873	37.78885
Jet amplitude	0.19172	0.18536
Jet width eta	0.36086	4.27056
jet width phi	0.29857	0.52430

- Fit parameters don't make sense

Different approach – 0-12% Au+Au 200



Raw



Subtracted

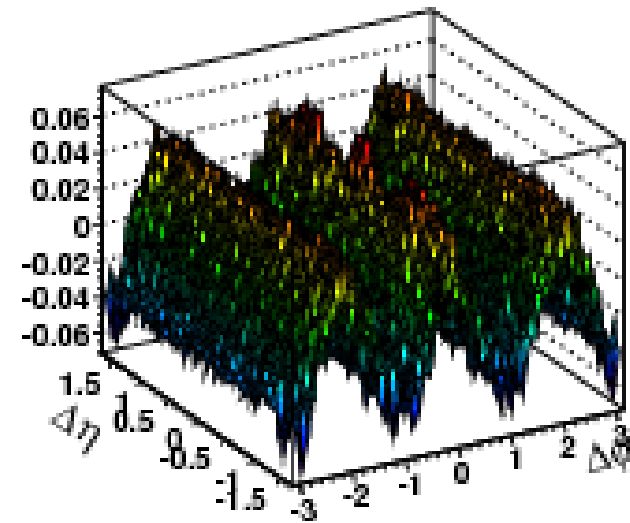
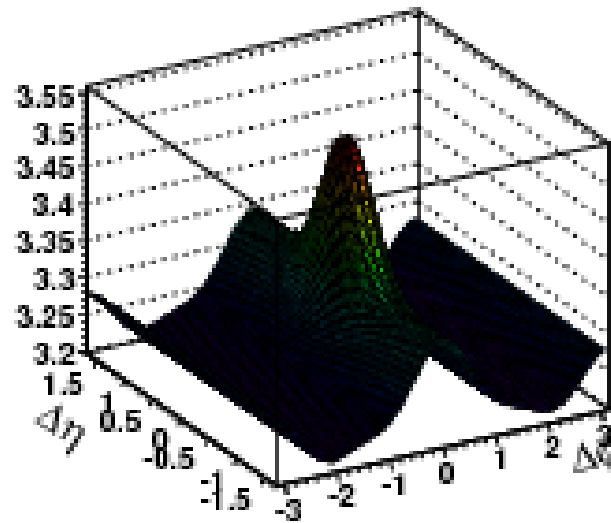
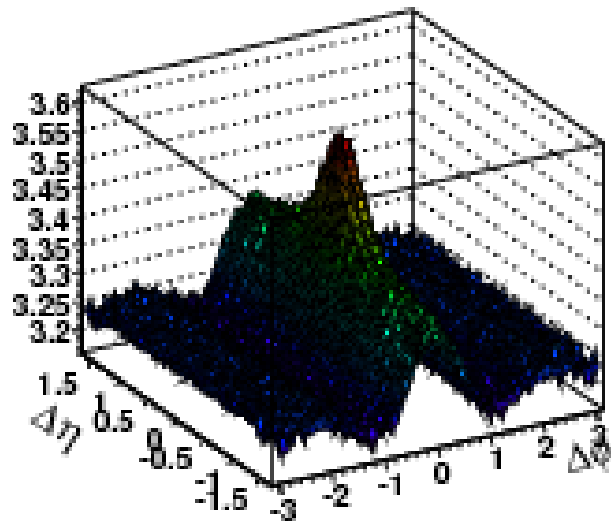
- Subtract non-jet-like correlation parts from fit of d+Au
- Except this is effectively just subtracting part of a Fourier series... and there is a question of what overall background level to use for the subtraction

Trying to subtract d+Au from Au+Au 0-12%

	Standard	After subtracting d+Au	d+Au
Background	3.34776	3.25551	0.09225
v1	0.00092	0.00464	-0.13004
(v2)^2	0.00811	0.00638	0.06943
(v3)^2	0.00499	0.00594	0.02850
(v4/v2)^2	0.24873	0.28437	0.13323
Jet amplitude	0.19172	0.19172	0.26890
Jet width eta	0.36086	0.36087	0.24691
jet width phi	0.29857	0.29857	0.25831

- This fit converges but I don't know what to conclude from it

Residuals



Global momentum conservation estimate: $c_1^{\Sigma p_T}(p_{T1}, y_1, j; p_{T2}, y_2, k) = -\frac{p_{T1}p_{T2}}{\langle \sum p_T^2 \rangle}$.

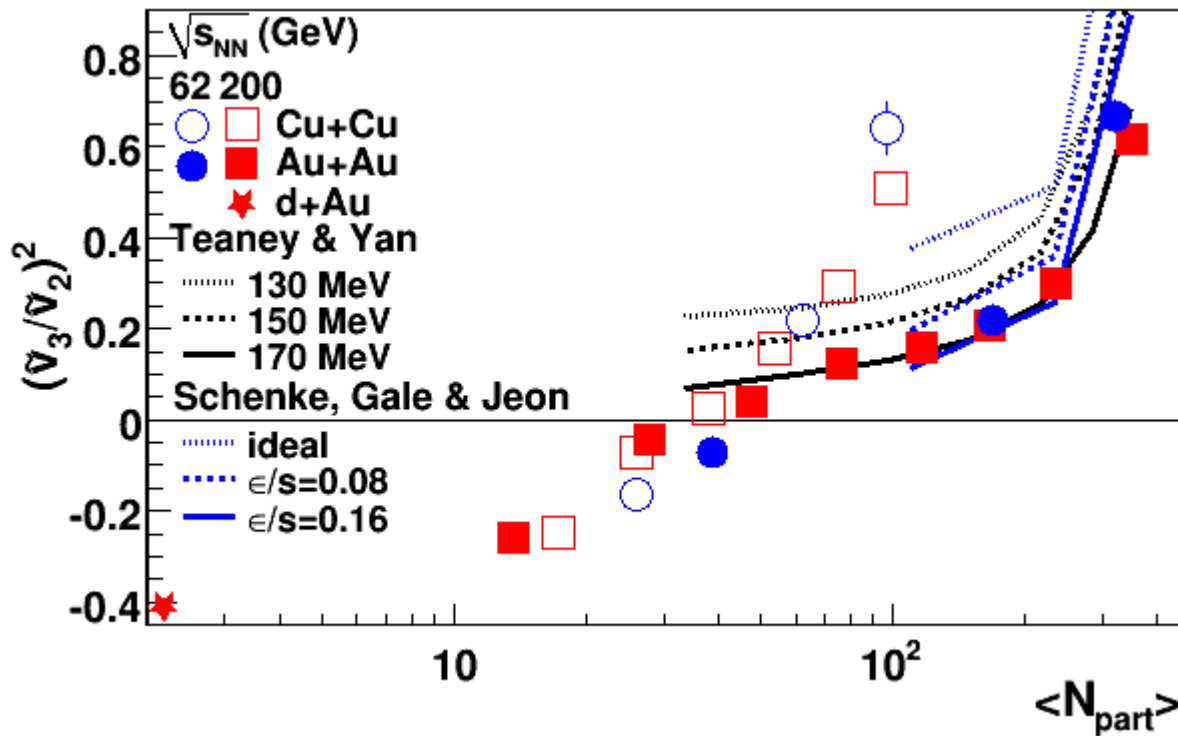
Centrality bin	c_1
0-5%	-0.020
5-10%	-0.023
10-20%	-0.032
20-30%	-0.048
30-40%	-0.073
40-50%	-0.12
50-60%	-0.20
60-70%	-0.42
70-80%	-0.94

$c1[i]=-(mpttrig[i]*mptass[i])/(2.0*ncharged[i]*mPt[i]*mPt[i]);$

Disclaimer: this is a “rough” estimate of c_1 as I do not have available at hand all quantities

- mPt taken from Fuqiang (thanks!) – slightly different centrality binning
- ncharged taken as the left edge of the centrality bin → c_1 is overestimated

v3 figure



- Found a problem with v3 fits:
 - v3 was restricted to positive values. The above is the result when not restricted. No change for data where $v_3^2 > 0$ before.
 - Analysis note updated
- Added d+Au to figure
- Added lines from Schenke
 - Appropriate title for theory TBD
 - With 6 theory lines there is no good way to draw the figure so it works in black & white
- Added a table with v_2^2 and v_3^2 to table in paper

Calculations from Petersen et al

